

POWER QUALITY DISTURBANCE CLASSIFICATION USING MACHINE LEARNING

BY INTERDEPARTMENTAL TEAM FROM
DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER ENGINEERING

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PROBLEM STATEMENT

- Increasing use of power electronic devices and nonlinear loads causes power quality disturbances such as voltage sag, voltage swell, harmonics, and interruptions.
- These disturbances can lead to equipment malfunction, reduced efficiency, production downtime, and higher maintenance costs.
- Conventional monitoring methods are manual, threshold-based, and not suitable for intelligent real-time classification.
- Testing on a 230 V AC supply poses safety risks and is not ideal for laboratory experimentation.
- Therefore, there is a need for a safe, automated, and AI-based system capable of accurately detecting and classifying power quality disturbances in real time.
- This project addresses the problem by developing a 12 V hardware prototype integrated with a machine learning model for automatic disturbance generation, classification, and experimental validation.

PROJECT OBJECTIVES

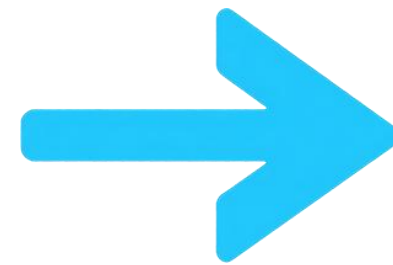
- Develop a Machine Learning model for power quality disturbance classification.
- Train the model using a Power Quality Disturbance (PQD) dataset.
- Design a safe 12 V automatic hardware prototype.
- Generate Normal, Voltage Sag, Voltage Swell, Interruption, and Harmonics automatically.
- Extract electrical features and perform real-time disturbance classification.
- Validate AI predictions through experimental hardware testing.

**THE WHOLE PROJECT IN ONE LINE IS
TO DEVELOP A SAFE, INTELLIGENT, AND REAL-TIME SYSTEM FOR
AUTOMATIC POWER QUALITY DISTURBANCE CLASSIFICATION
USING MACHINE LEARNING AND A 12 V LABORATORY PROTOTYPE.**

OVERALL PROJECT WORKFLOW

PHASE 1: Data & Model

- ▶ Dataset Collection
- ▶ Preprocessing & Normalization
- ▶ Feature Extraction (RMS, THD, Crest Factor)
- ▶ ML Model Training (CNN / Random Forest)
- ▶ TensorFlow Lite Conversion



PHASE 2: Hardware

- ▶ ESP32 Prototype Setup
- ▶ Relay Module Configuration
- ▶ Automatic Disturbance Generation
- ▶ Voltage & Current Measurement
- ▶ Feature Calculation

PHASE 3: Validation

- ▶ Real-Time AI Classification
- ▶ Prediction Output on LCD Display
- ▶ Confidence Score Evaluation
- ▶ Experimental Validation
- ▶ Performance Analysis



DATASET & AI DEVELOPMENT

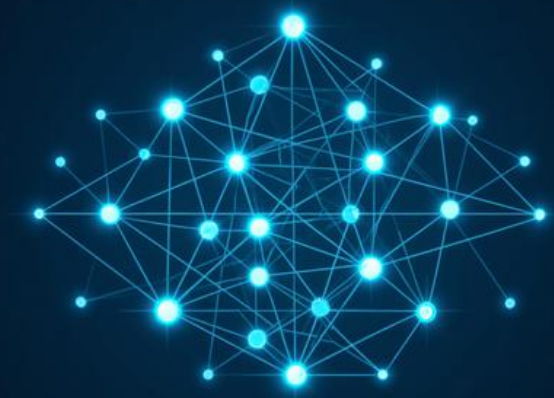
Dataset Features: RMS Voltage, Peak Voltage, Crest Factor, THD, Disturbance Duration, Dominant Frequency, System Frequency, Signal-to-Noise Ratio. **Target Classes:** Normal, Voltage Sag, Voltage Swell, Interruption, Harmonics.

AI Pipeline: Data preprocessing → Label encoding → Feature normalization → CNN/ML classifier training → Model evaluation using accuracy, precision, recall, and F1-score metrics.

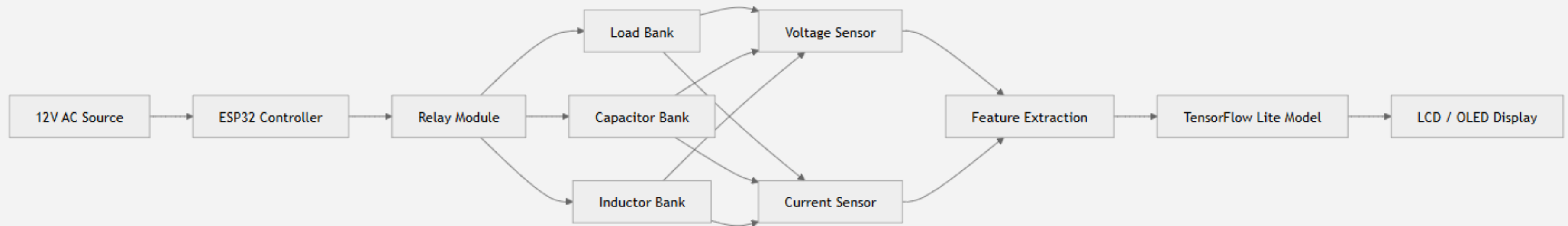
Model Deployment: Trained classifier converted to TensorFlow Lite format for embedded deployment on ESP32 microcontroller, enabling real-time on-device power quality disturbance classification.

Elected Tabl/es

	Voltage	THD	CP-D	Crest Factor	Frequency	Frequency
RMS Voltage	55-5.3	26.4.3	1.5.6	5.5.1	5.2.1	5.6.5
Exepialce	5.6	55-7.3	5	5	5	6
Inizpiance	14.33	1.23	8	75.7	8.8.5	9.9.3
BMV atrrands	324	8.9	92	45	71	5.4.3
Crest Factor	2/6	272	4	5	4	2
Crest Factory	0.6	5	4	4	1	5
Aultvas	2.7	808	79	5	5	200
New Attrones	2.2.8	14.71.4	95	118	01.K	10.62.5
Fish cention	1.8.8	35.7.73	3604.15	8.234.18	3.0/4.15	10:24.15



12V AUTOMATIC DISTURBANCE GENERATION PROTOTYPE



Key Features

- Automatic relay-controlled disturbance generation
- Safe 12 V operation (no 230 V mains)
- Generates Normal, Sag, Swell, Interruption & Harmonics
- Real-time voltage and current monitoring
- Supports on-device TensorFlow Lite inference
- Instant display of disturbance type and confidence

HARDWARE COMPONENTS EXPECTED TO BE USED

Component	Purpose
12 V AC Transformer / Power Supply	Provides a safe low-voltage AC source for the prototype
ESP32 Development Board	Controls relay switching, data acquisition, and AI inference
4/8-Channel Relay Module	Automatically switches circuit configurations to generate disturbances
Voltage Sensor Module	Measures the output voltage waveform
Current Sensor (ACS712 or equivalent)	Measures load current for feature extraction
Load Bank (Resistive Loads)	Creates different loading conditions and voltage sag/interruption
Capacitor Bank	Generates voltage swell and reactive power variations
Inductor Bank	Produces inductive loading and harmonic-related effects
LCD/OLED Display (I ² C)	Displays predicted disturbance, confidence, and system status
5 V Power Supply Module	Powers the ESP32, sensors, and relay module
Connecting Wires & PCB/Breadboard	Hardware interconnections and circuit assembly
Protection Components (Fuse, Terminal Blocks, Connectors)	Ensures safe operation and reliable connections

POWER QUALITY DISTURBANCE GENERATION

Normal Signal

Standard sinusoidal waveform at rated voltage and frequency. Serves as the baseline reference for all disturbance comparisons.

Voltage Sag

Temporary reduction in RMS voltage (10%–90% of nominal) lasting 0.5 cycles to 1 minute. Caused by faults or large load switching.



Voltage Swell

Temporary increase in RMS voltage (110%–180% of nominal). Caused by sudden load disconnection or capacitor bank switching.

Interruption

Complete loss of supply voltage (below 10% of nominal) for a duration exceeding 0.5 cycles. Caused by faults or protective device operation.

Harmonics

Distorted waveform containing multiple frequency components (integer multiples of 50 Hz). Caused by non-linear loads and power electronics.



EXPECTED OUTCOMES



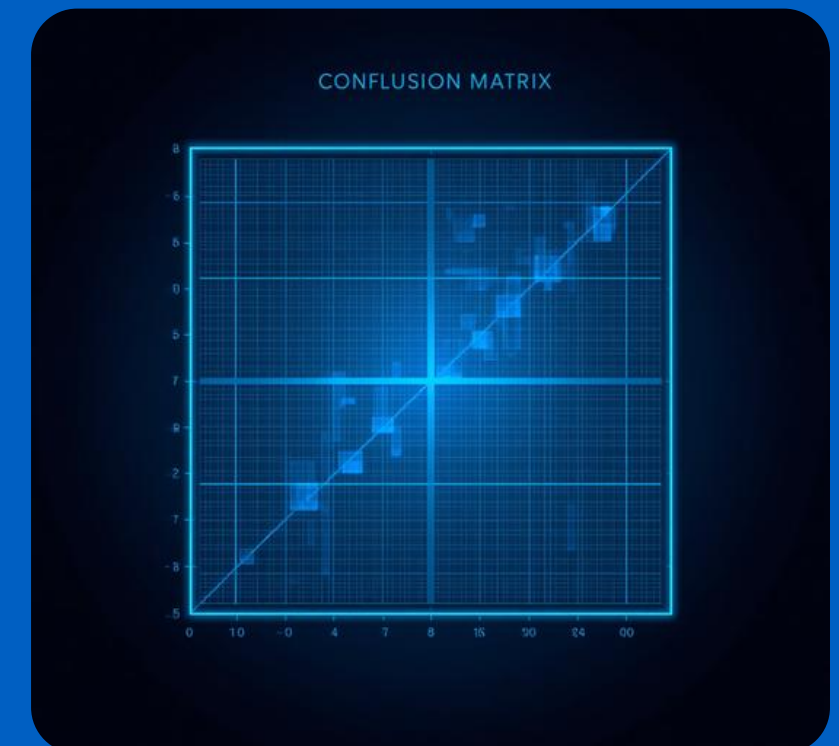
High Classification Accuracy & Real-Time Detection

The ML model is expected to achieve $>95\%$ accuracy across all five disturbance classes. TensorFlow Lite deployment on ESP32 enables real-time inference with low latency, ensuring instant disturbance identification without cloud dependency.



Low-Cost Prototype & Industry-Oriented Design

The 12V automatic prototype demonstrates a fully functional, safe, and affordable hardware solution. Relay-controlled disturbance generation and embedded AI classification make it scalable for industrial power monitoring applications.



TensorFlow Lite Deployment & Research Publication Readiness

Successful edge AI deployment on an embedded microcontroller validates the system for IoT-based power quality monitoring. The methodology, dataset, and results are structured for submission to IEEE or equivalent electrical engineering conferences.

APPLICATIONS



Smart Grid

Intelligent monitoring and automated fault detection in modern power grids.

Industrial Automation

Real-time PQ disturbance detection to protect sensitive industrial equipment.

Power Distribution

Continuous quality monitoring across distribution networks for reliability.



Renewable Energy

Ensuring stable integration of solar and wind sources into the grid.

Electric Vehicles

Protecting EV charging infrastructure from voltage sags and harmonics.

Microgrids

Automated disturbance classification for isolated and hybrid microgrid systems.

IoT-Based Monitoring

Remote PQ surveillance via connected sensors and cloud dashboards.

Predictive Maintenance

Early fault identification to prevent equipment failure and reduce downtime.

SCADA Integration

Seamless compatibility with industrial supervisory control and data systems.



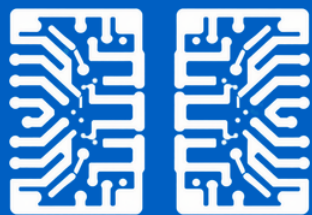
FUTURE ENHANCEMENTS



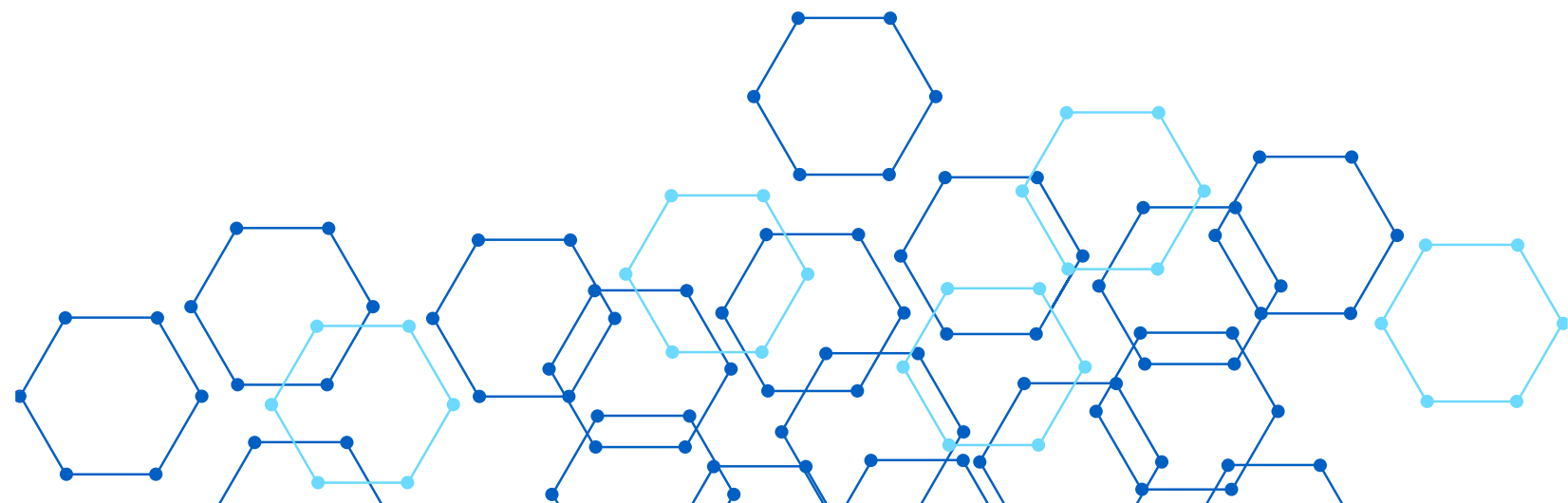
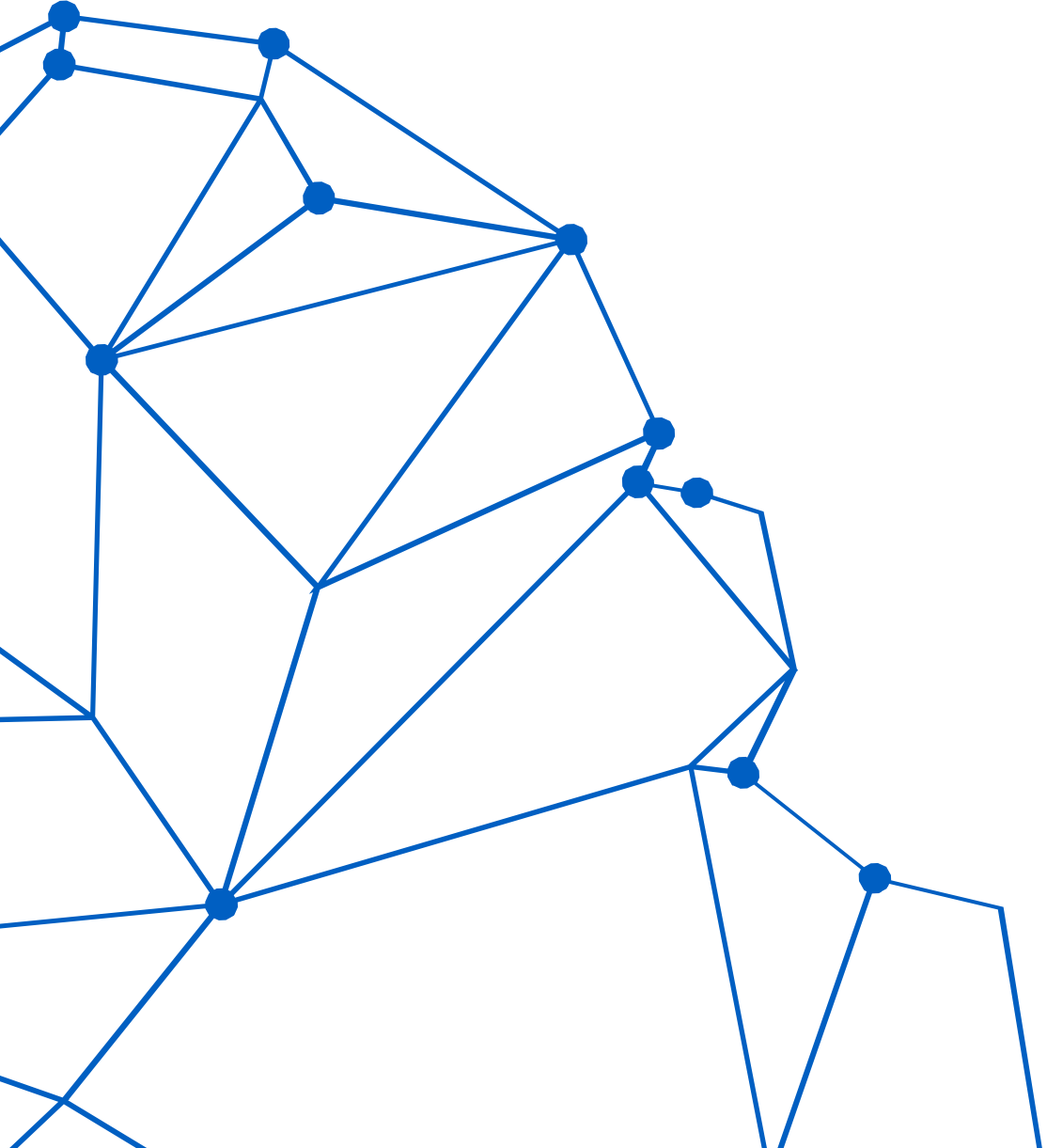
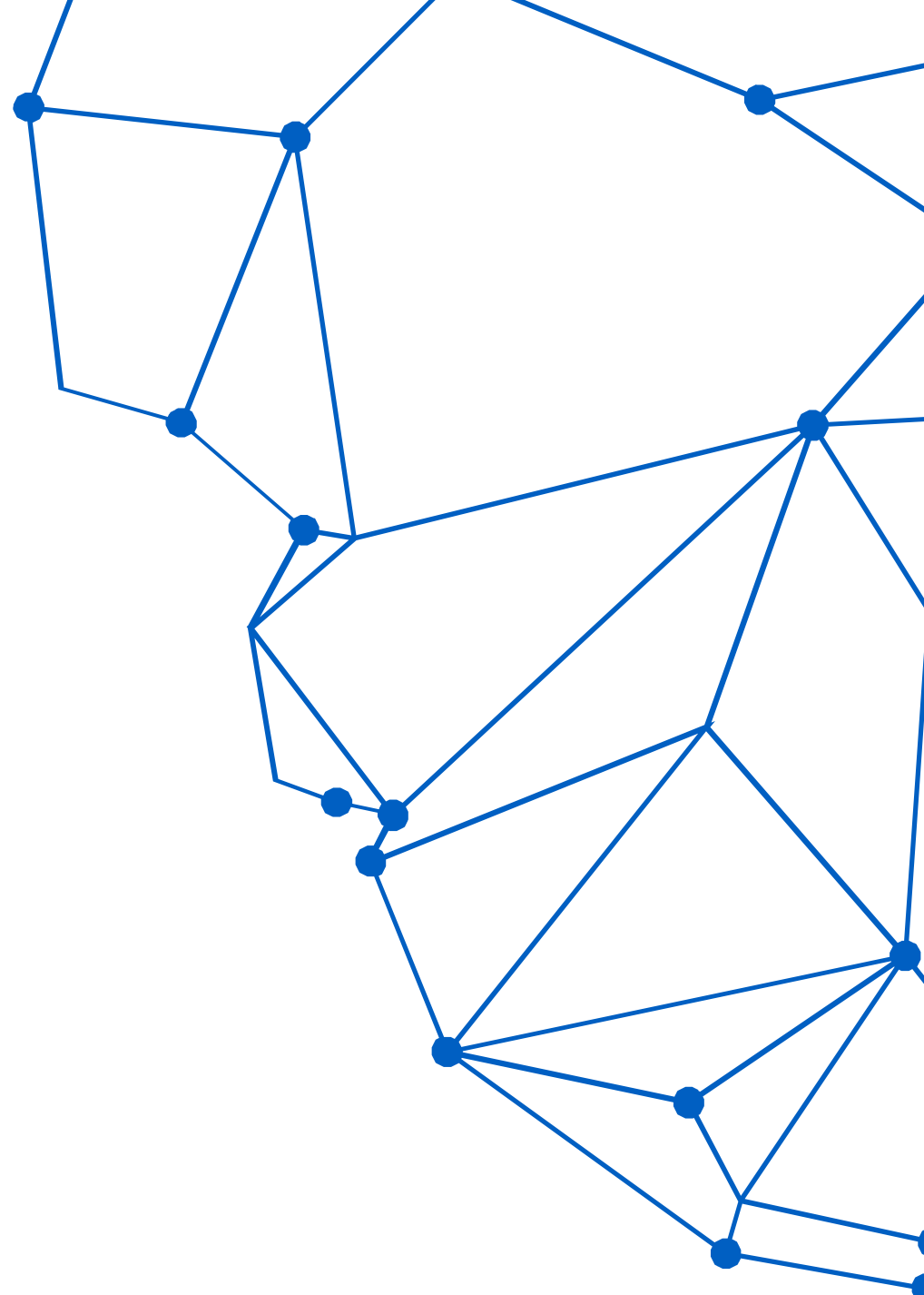
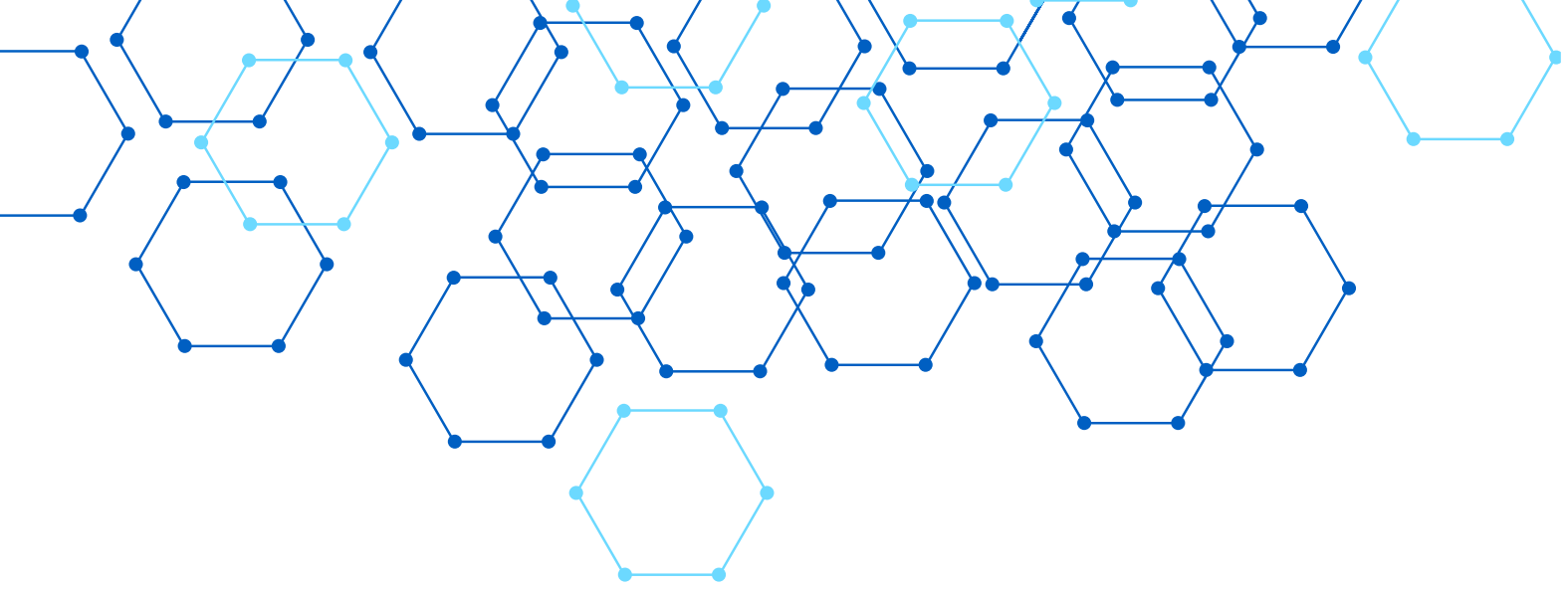
Extended Disturbance Classes & Deep Learning Optimization: Expand classification to support all eight disturbance classes. Integrate advanced deep learning architectures (CNN, LSTM) to improve accuracy, reduce inference time, and enhance model robustness on embedded hardware.



Cloud Monitoring, IoT Dashboard & Edge AI: Enable remote real-time monitoring via cloud-connected IoT dashboard. Implement Edge AI for faster local inference with reduced latency, and integrate SCADA compatibility for seamless industrial system deployment.



Digital Twin & High-Voltage Industrial Testing: Develop a Digital Twin simulation framework for virtual prototyping and analysis. Scale the prototype to high-voltage industrial environments to validate real-world performance and research publication readiness.



**THANK YOU!
QUESTIONS?**